

CSE24031

```
% Generate random square matrices A, B, and C
A = randi([-10, 10], 4, 4);
B = randi([-10, 10], 10, 10);
C = randi([-10, 10], 15, 15);
% Find their eigenvalues and eigenvectors
[eigenvectors, eigenvalues] = eig(A);
[eigenvectors, eigenvalues] = eig(B);
[eigenvectors, eigenvalues] = eig(C);
eigA = eig(A);
eigB = eig(B);
eigC = eig(C);
% Calculate trace and sum of eigenvalues
traceA = trace(A)
```

```
traceA =
20
```

```
sumEigA = sum(eigA)
```

```
sumEigA =
20.0000
```

```
traceB = trace(B)
```

```
traceB =
-39
```

```
sumEigB = sum(eigB)
```

```
sumEigB =
-39.0000
```

```
traceC = trace(C)
```

```
traceC =
-1
```

```
sumEigC = sum(eigC)
```

```
sumEigC =
-1.0000
```

```
% Check if trace equals sum of eigenvalues for all matrices.
if round(traceA) == round(sumEigA) && round(traceB) == round(sumEigB) &&
round(traceC) == round(sumEigC)
    disp('Trace equals sum of eigenvalues for all matrices.');
```

```
else
```

```
    disp('Trace does not equal sum of eigenvalues for all matrices.');
```

```
end
```

```
Trace equals sum of eigenvalues for all matrices.
```

```
% Calculate determinant as product of eigenvalues
D1 = det(A)
```

```
D1 =
```

-6.4980e+03

```
P1 = prod(eigA)
```

```
P1 =  
-6.4980e+03 - 3.5190e-13i
```

```
D2 = det(B)
```

```
D2 =  
-1.6930e+11
```

```
P2= prod(eigB)
```

```
P2 =  
-1.6930e+11
```

```
D3 = det(C)
```

```
D3 =  
3.8321e+17
```

```
P3=prod(eigC)
```

```
P3 =  
3.8321e+17
```

```
% Check if determinant equals product of eigenvalues for all matrices.  
if round(D1) == round(P1) && round(D2) == round(P2) && round(D3) == round(P3)  
    disp('Determinant equals product of eigenvalues for all matrices.');
```

```
else  
    disp('Determinant does not equal product of eigenvalues for all matrices.');
```

```
end
```

Determinant does not equal product of eigenvalues for all matrices.

```
eigenvalues_A = eig(A);  
eigenvalues_B = eig(B);  
eigenvalues_C = eig(C);  
for M = {eigenvalues_A, eigenvalues_B, eigenvalues_C}  
    if isequal(sort(conj(M{1})(imag(M{1})~=0))), sort(M{1})(imag(M{1})~=0)))  
        disp('Complex eigenvalues appear in conjugate pairs.');
```

```
    else  
        disp('Complex eigenvalues do NOT appear in conjugate pairs.');
```

```
    end  
end
```

Complex eigenvalues appear in conjugate pairs.
Complex eigenvalues appear in conjugate pairs.
Complex eigenvalues appear in conjugate pairs.

```
% Find eigenvalues of 5A, B/3 and C^-1. Compare them with eigenvalues of *, *  
and * respectively.  
P= 5*A;  
P1= eig(P)
```

```
P1 = 4x1 complex  
-34.0704 + 0.0000i  
51.5585 +34.5359i  
51.5585 -34.5359i
```

30.9534 + 0.0000i

```
Q=(B/3);  
Q1=eig(Q)
```

```
Q1 = 10×1 complex  
4.9545 + 0.0000i  
-7.0455 + 0.0000i  
-4.0895 + 5.8558i  
-4.0895 - 5.8558i  
1.1474 + 5.2114i  
1.1474 - 5.2114i  
-1.2356 + 4.0456i  
-1.2356 - 4.0456i  
-1.2768 + 1.2368i  
-1.2768 - 1.2368i
```

```
R=C^(-1);  
R1=eig(R)
```

```
R1 = 15×1 complex  
0.0577 + 0.1019i  
0.0577 - 0.1019i  
0.0704 + 0.0000i  
0.0631 + 0.0173i  
0.0631 - 0.0173i  
-0.0574 + 0.0701i  
-0.0574 - 0.0701i  
0.0239 + 0.0446i  
0.0239 - 0.0446i  
-0.0114 + 0.0637i  
⋮
```

```
[eigVecP, ~] = eig(P)
```

```
eigVecP = 4×4 complex  
-0.0231 + 0.0000i -0.6425 + 0.0000i -0.6425 + 0.0000i 0.3028 + 0.0000i  
0.8398 + 0.0000i -0.4932 - 0.1093i -0.4932 + 0.1093i 0.2149 + 0.0000i  
0.4088 + 0.0000i 0.0113 + 0.3759i 0.0113 - 0.3759i -0.2213 + 0.0000i  
0.3566 + 0.0000i -0.3912 + 0.1938i -0.3912 - 0.1938i 0.9018 + 0.0000i
```

```
[eigVecQ, ~] = eig(Q)
```

```
eigVecQ = 10×10 complex  
0.3782 + 0.0000i 0.7633 + 0.0000i -0.0274 + 0.0525i -0.0274 - 0.0525i ...  
0.0288 + 0.0000i -0.1916 + 0.0000i 0.0533 - 0.3091i 0.0533 + 0.3091i  
0.1995 + 0.0000i -0.1839 + 0.0000i -0.5263 + 0.0000i -0.5263 + 0.0000i  
0.5035 + 0.0000i 0.0140 + 0.0000i 0.3748 - 0.3038i 0.3748 + 0.3038i  
0.1301 + 0.0000i 0.1879 + 0.0000i -0.0947 - 0.2472i -0.0947 + 0.2472i  
-0.3155 + 0.0000i -0.0725 + 0.0000i -0.2619 - 0.2168i -0.2619 + 0.2168i  
-0.3263 + 0.0000i 0.1169 + 0.0000i -0.0218 - 0.1018i -0.0218 + 0.1018i  
-0.4824 + 0.0000i 0.3603 + 0.0000i 0.0614 - 0.2975i 0.0614 + 0.2975i  
0.3186 + 0.0000i -0.1216 + 0.0000i -0.2330 - 0.0287i -0.2330 + 0.0287i  
-0.0756 + 0.0000i 0.3844 + 0.0000i 0.0794 + 0.1953i 0.0794 - 0.1953i
```

```
[eigVecR, ~] = eig(R)
```

```
eigVecR = 15×15 complex  
0.0635 + 0.0451i 0.0635 - 0.0451i 0.1222 + 0.0000i 0.1061 - 0.2380i ...  
-0.1551 - 0.0447i -0.1551 + 0.0447i 0.4800 + 0.0000i -0.2043 - 0.2561i  
-0.3162 + 0.2654i -0.3162 - 0.2654i -0.3564 + 0.0000i 0.3017 + 0.0685i  
-0.0655 + 0.0079i -0.0655 - 0.0079i 0.2775 + 0.0000i -0.3003 - 0.0248i  
-0.2005 - 0.0194i -0.2005 + 0.0194i 0.0264 + 0.0000i -0.0588 + 0.1142i  
0.5187 + 0.0000i 0.5187 + 0.0000i 0.2921 + 0.0000i -0.3309 + 0.0598i  
-0.1922 - 0.0586i -0.1922 + 0.0586i -0.4512 + 0.0000i 0.3375 + 0.0000i
```

```
-0.1846 - 0.0241i -0.1846 + 0.0241i -0.1333 + 0.0000i 0.1751 - 0.0467i
-0.0934 - 0.0664i -0.0934 + 0.0664i -0.1754 + 0.0000i 0.2707 - 0.0213i
0.2092 + 0.1296i 0.2092 - 0.1296i 0.1042 + 0.0000i 0.0817 - 0.1374i
⋮
```

```
% From A, using A = A + A' generate a symmetric matrix. Find the eigenvalues
and eigenvectors of S.
S=A+A'
```

```
S = 4x4
    16    15     2     4
    15     0    -17    -6
     2    -17    18    -5
     4     -6    -5     6
```

```
[eigenvectors, eigenvalues] = eig(S)
```

```
eigenvectors = 4x4
   -0.3836    0.1219    0.7803   -0.4786
    0.7614   -0.3008    0.0718   -0.5698
    0.4111   -0.0696    0.6204    0.6643
    0.3227    0.9433   -0.0322   -0.0708
eigenvalues = 4x4
   -19.2792     0         0         0
         0    8.7994     0         0
         0     0   18.8062     0
         0     0     0   31.6735
```

2) Generate a random integer square matrix A of order 3.

```
% a) Find the eigenvalues, right eigenvectors, and left eigenvectors of this
matrix.
clc
A = randi([-5,5],3);
[R,L]=eig(A)
```

```
R = 3x3 complex
   -0.3043 - 0.2621i   -0.3043 + 0.2621i    0.1485 + 0.0000i
    0.7184 + 0.0000i    0.7184 + 0.0000i   -0.0986 + 0.0000i
    0.5625 + 0.0782i    0.5625 - 0.0782i    0.9840 + 0.0000i
L = 3x3 complex
    0.4775 + 1.5681i    0.0000 + 0.0000i    0.0000 + 0.0000i
    0.0000 + 0.0000i    0.4775 - 1.5681i    0.0000 + 0.0000i
    0.0000 + 0.0000i    0.0000 + 0.0000i   -5.9550 + 0.0000i
```

```
eig(A)
```

```
ans = 3x1 complex
    0.4775 + 1.5681i
    0.4775 - 1.5681i
   -5.9550 + 0.0000i
```

```
% b) Find the characteristic polynomial for this matrix.
syms x;
eq1=charpoly(A, x)
```

```
eq1 = x3 + 5 x2 - 3 x + 16
```

```
% c) Replace x in the characteristic polynomial by the matrix A and evaluate the
value. What is the value obtained? Try this with other square matrices.
```

```
Eq2=subs(eq1,A)
```

Eq2 =

$$\begin{pmatrix} 38 & 38 & 23 \\ 44 & 34 & 19 \\ 31 & 38 & 31 \end{pmatrix}$$

3) Generate an integer matrix A of order 5×3. Find $\lambda_1, \lambda_2, \lambda_3$ and λ_4, λ_5 . Evaluate the eigenvalues of $\lambda_1, \lambda_2, \lambda_3$ and λ_4, λ_5 . Verify the result that “The non-zero eigenvalues of $\lambda_1, \lambda_2, \lambda_3$ and λ_4, λ_5 are the same”.

```
clc
A=randi([0,9],5,3);
rank(A);
A1=A*A'
```

```
A1 = 5×5
    133    110    125    120    39
    110     93     97     92    34
    125     97    138    138    31
    120     92    138    149    28
     39     34     31     28    13
```

```
A2=A'*A
```

```
A2 = 3×3
    267    151    180
    151     94     92
    180     92    165
```

```
% Compute Eigenvalues for both the matrices
E_A1=eig(A1)
```

```
E_A1 = 5×1
    -0.0000
    -0.0000
     3.9405
    39.2287
   482.8308
```

```
E_A2=eig(A2)
```

```
E_A2 = 3×1
     3.9405
    39.2287
   482.8308
```

```
% Remove near-zero eigenvalues for comparison
E_A1_non_zero = E_A1(E_A1 > 1e-10)
```

```
E_A1_non_zero = 3×1
     3.9405
    39.2287
   482.8308
```

```
E_A2_non_zero = E_A2(E_A2 > 1e-10)
```

```
E_A2_non_zero = 3×1
     3.9405
    39.2287
```

```
% Compare non-zero eigenvalues
if all(abs(E_A1_non_zero - E_A2_non_zero) < 1e-10)
    disp('The non-zero eigenvalues of A*A'' and A''*A are the same.');
```

```
else
    disp('The non-zero eigenvalues of A*A'' and A''*A are NOT the same.');
```

```
end
```

The non-zero eigenvalues of A*A' and A'*A are the same.

4) Generate a matrix with eigenvalues as 3,5,7,11 and eigenvectors as [1,1,2,2], [1,0,1,0], [2,1,2,0] and [0,1,1,0].

```
clc
% Enter the given eigenvalues
Ld1 = 3;
Ld2 = 5;
Ld3 = 7;
Ld4= 11;
% Enter the given eigenvectors
Ev1 = [1;1;2;2];
Ev2 = [1;0;1;0];
Ev3 = [2;1;2;0];
Ev4=[0;1;1;0];
V = [Ev1,Ev2,Ev3,Ev4];
D = diag([Ld1,Ld2,Ld3,Ld4]);
A = V*D*inv(V)
```

```
A = 4x4
     9     4    -4    -1
    -4     7     4    -4
    -2     4     7    -5
     0     0     0     3
```

5) Using MATLAB generate a 5 × 5 matrix of rank 2.

```
A = randi([0 9], 5, 2)*randi([0 9], 2,5);
B=A+A'
```

```
B = 5x5
    224    39   131   137   151
     39     0    54    61     5
    131    54    44    31   107
    137    61    31    14   122
    151     5   107   122    90
```

```
rankB=rank(B)
```

```
rankB =
4
```

```
% b) Find the eigenvalues and eigenvectors of using MATLAB.
[eigenvectors, eigenvalues] = eig(B)
```

```
eigenvectors = 5x5
    -0.1733    0.6992    0.0454   -0.0787    0.6876
```

```

-0.4217   -0.1382   -0.2801    0.8382    0.1487
 0.4280   -0.2628    0.7039    0.3419    0.3678
 0.6425   -0.1613   -0.6510    0.0134    0.3705
-0.4430   -0.6300   -0.0151   -0.4173    0.4822
eigenvalues = 5x5
-126.4686         0         0         0         0
         0   -0.6114         0         0         0
         0         0   -0.0000         0         0
         0         0         0   16.8505         0
         0         0         0         0  482.2296

```

% c) Are their complex eigenvalues? (Check the same for more symmetric matrices). Explain why or why not?

% Check if there are complex eigenvalues

```

if isreal(eigenvalues)
    disp('All eigenvalues are real.');
```

else

```

    disp('There are complex eigenvalues.');
```

end

All eigenvalues are real.

% To count how many eigenvalues are zero

```
zeroEigenvalues = sum(abs(diag(eigenvalues)) < 1e-6)
```

```

zeroEigenvalues =
1

```

```
expected_Zero_Eigenvalues = size(B,1) - rankB
```

```

expected_Zero_Eigenvalues =
1

```

% To check whether there is a relation between the number of zero eigenvalues of B.

```

if zeroEigenvalues == expected_Zero_Eigenvalues
    disp('The number of zero eigenvalues matches the expected relation with the rank
of B.');
```

else

```

    disp('The number of zero eigenvalues does not match the expected relation with
the rank of B.');
```

end

The number of zero eigenvalues matches the expected relation with the rank of B.

% e) Verify if the eigenvectors of are orthogonal.

% Check if eigenvectors are orthogonal.

```
orthogonalCheck = eigenvectors' * eigenvectors;
```

% Check if eigenvectors' * eigenvectors is close to identity matrix

```
if norm(orthogonalCheck - eye(size(B,1))) < 1e-10
```

```
    disp('Eigenvectors are orthogonal.');
```

else

```
    disp('Eigenvectors are not orthogonal.');
```

end

Eigenvectors are orthogonal.

6. Using MATLAB, generate a 10×5 matrix $\blacklozenge$ of rank 3.

```
% a. Obtain a symmetric matrix  $S_1 = A * A'$  and  $S_2 = A * A'$ 
```

```
clc
```

```
A = randi([0 9], 10, 3)*randi([0 9], 3,5)
```

```
A = 10x5
```

9	33	42	6	25
71	64	135	67	54
87	93	180	78	91
42	87	129	33	77
87	78	165	81	72
51	105	156	42	83
78	79	157	71	75
46	51	97	43	39
49	72	121	43	60
31	44	75	29	28

```
S1 = A'*A
```

```
S1 = 5x5
```

36187	42127	78314	32694	37159
42127	54474	96601	37513	47240
78314	96601	174915	70207	84399
32694	37513	70207	29603	33119
37159	47240	84399	33119	41354

```
S2 = A*A'
```

```
S2 = 10x10
```

3595	10173	14155	10790	12573	12803 ...
10173	34767	46569	32334	42759	38697
14155	46569	62983	44546	57393	53111
10790	32334	44546	32992	39942	39178
12573	42759	57393	39942	52623	47745
12803	38697	53111	39178	47745	46615
12204	40596	54756	38520	50004	45972
7404	24612	33108	23304	30276	27876
9657	30543	41553	29970	37647	35721
5755	18597	25099	17918	22869	21443

```
% b. Find the rank of  $S_1$  and  $S_2$ .
```

```
rank(S1)
```

```
ans =  
3
```

```
rank(S2)
```

```
ans =  
3
```

```
% c. Find the eigenvalues of  $S_1$  and  $S_2$ .
```

```
EvalS1 = eig(S1)
```

```
EvalS1 = 5x1
```

```
105 x
```

-0.0000
0.0000
0.0021
0.0406
3.3226

```
EvalS2 = eig(S2)
```

```
EvalS2 = 10x1
```



```

105 ×
-0.0000
-0.0000
-0.0000
0.0000
0.0000
0.0000
0.0000
0.0021
0.0406
3.3226

```

7. Using MATLAB, generate a 7×7 matrix $\mathbf{A}$ of rank 2.

% a. Obtain a skew-symmetric matrix $\mathbf{B} = \mathbf{A} - \mathbf{A}'$ and find the rank of $\mathbf{B}$.

```

clc
A = randi([0 9],7,2)*randi([0 9],2,7);
B = A - A'

```

```

B = 7×7
    0    93    27    36   111    45    18
   -93     0   -24     4    -4   -22   -79
   -27    24     0    14    32    12   -11
   -36    -4   -14     0    -1   -14   -31
  -111     4   -32     1     0   -39  -104
   -45    22   -12    14    39     0   -26
   -18    79    11    31   104    26     0

```

```
rank(B)
```

```

ans =
4

```

% b. Find the eigenvalues and eigenvectors of $\mathbf{B}$ using MATLAB.

```
[eigenvectors, eigenvalues] = eig(B)
```

```

eigenvectors = 7×7 complex
-0.5127 + 0.0008i  -0.5127 - 0.0008i  -0.3303 + 0.0766i  -0.3303 - 0.0766i ...
 0.0682 - 0.4010i  0.0682 + 0.4010i  -0.2919 - 0.2149i  -0.2919 + 0.2149i
-0.1344 - 0.1002i  -0.1344 + 0.1002i  -0.1470 - 0.4073i  -0.1470 + 0.4073i
 0.0406 - 0.1608i  0.0406 + 0.1608i  0.2095 - 0.0343i  0.2095 + 0.0343i
 0.0822 - 0.5096i  0.0822 + 0.5096i  -0.0047 + 0.3603i  -0.0047 - 0.3603i
-0.1252 - 0.1883i  -0.1252 + 0.1883i  0.3921 - 0.3509i  0.3921 + 0.3509i
-0.4363 - 0.0898i  -0.4363 + 0.0898i  0.2940 + 0.1672i  0.2940 - 0.1672i
eigenvalues = 7×7 complex
102 ×
 0.0000 + 2.1933i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i ...
 0.0000 + 0.0000i  0.0000 - 2.1933i  0.0000 + 0.0000i  0.0000 + 0.0000i
 0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.1514i  0.0000 + 0.0000i
 0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 - 0.1514i
 0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i
 0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i
 0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i  0.0000 + 0.0000i

```

% Check if eigenvalues are real (they should be imaginary for skew-symmetric matrices).

```

if all(isreal(diag(eigenvalues)))
disp('All eigenvalues are real.')
else
disp('Not all eigenvalues are real. There are imaginary eigenvalues.')
end

```

Not all eigenvalues are real. There are imaginary eigenvalues.

```
% d) Do you have one eigenvalue as zero? Can you explain why?
% Check if there s a one or zero eigenvalue.
has_zero_eigenvalue = any(abs(diag(eigenvalues)) < 1e-10)
```

```
has_zero_eigenvalue = logical
1
```

```
% e) In a similar manner generate a skew-symmetric matrix from an 8× 8 matrix.
Does C have a zero eigenvalue? Why or why not?
% Create an 8*8 matrix and generate a skew symmetric matrix C.
C = rand(8, 8) - rand(8, 8)'
```

```
C = 8×8
    0.0518   -0.0362    0.6042   -0.4236    0.7835   -0.3856   -0.1351   -0.5365
    0.0707    0.2972    0.0220    0.3112    0.2439   -0.7444   -0.4104    0.3570
    0.5941    0.5916    0.1298    0.3616    0.5061    0.5589    0.3670    0.2445
   -0.0726    0.5241   -0.1802   -0.0384    0.6917    0.2822    0.1202    0.0121
   -0.0593   -0.3429   -0.0584    0.7017    0.6505    0.5444    0.0860   -0.2164
    0.1727    0.0582   -0.2600    0.3432   -0.8908   -0.1399    0.4451   -0.0541
   -0.8472   -0.5909   -0.2867    0.0591   -0.1138   -0.5132   -0.1313    0.1280
   -0.1139    0.0838    0.6714    0.5560    0.1871   -0.0137    0.0486   -0.4272
```

```
% Find the eigenvalues of C and check if it has a zero eigenvalue.
eigenvalues_C = eig(C)
```

```
eigenvalues_C = 8×1 complex
   -0.9222 + 0.1679i
   -0.9222 - 0.1679i
   -0.2619 + 0.8422i
   -0.2619 - 0.8422i
    0.9060 + 0.6449i
    0.9060 - 0.6449i
    0.4744 + 0.0791i
    0.4744 - 0.0791i
```

```
has_zero_eigenvalue_C = any(abs(eigenvalues_C) < 1e-10)
```

```
has_zero_eigenvalue_C = logical
0
```

```
if has_zero_eigenvalue_C
    disp('C has no zero eigenvalue.')
else
    disp('C has a zero eigenvalue.')
end
```

C has a zero eigenvalue.

8. Generate a matrix that will transform the points on a circular membrane to an elliptic membrane with a stretch factor of 5 and 9 along the directions [1,1] and [1, -1] respectively.

```
clc
% Eigenvector matrix for directions [1,1] and [1,-1]
V = (1/sqrt(2)) * [1 1; 1 -1];
% Diagonal scaling matrix with stretch factors.
D = [5 0; 0 9];
```

```
% Transformation matrix.
```

```
T = V * D * inv(V)
```

```
T = 2x2
```

```
    7    -2  
   -2     7
```